Advanced Control Systems

Assignment 4: Recursive Newton-Euler Formulation, Dynamic Regressor, Base Parameters, and Least-Squares Identification for a 3-DOF PRR Manipulator

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Abstract

This report addresses the Recursive Newton-Euler (RNE) formulation, linearity in dynamic parameters, base parameter reduction, and Least-Squares (LS) parameter identification for a spatial 3-DOF PRR (Prismatic-Revolute-Revolute) robotic manipulator defined by URDF specification (`00001_PRR_zzx FILIPPO SEGALLA.urdf`). First, the $O(n)$ recursive Newton-Euler vector algorithm (comprising forward velocity/acceleration propagation and backward force/torque propagation) is formulated and numerically verified against the Euler-Lagrange equations from Assignment #3. Second, the dynamic model is restructured into linear form $\boldsymbol{\tau} = \mathbf{Y}(q, \dot{q}, \ddot{q})\boldsymbol{\Theta}$. A linear dependence analysis reduces the initial parameter set to a minimal 5-element base parameter vector $\bar{\boldsymbol{\Theta}}$ with minimal regressor matrix $\mathbf{X}(q, \dot{q}, \ddot{q})$. Finally, dynamic parameter identification is executed using a persistent excitation trajectory, yielding exact numerical identification (error $< 10^{-15}$) with a well-conditioned stacked regressor matrix.

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1 Recursive Newton-Euler (RNE) Formulation

1.1 Theoretical Framework & Algorithmic Structure

What is being done: We implement the Recursive Newton-Euler (RNE) algorithm to compute inverse dynamics $\boldsymbol{\tau} = \text{RNE}(q, \dot{q}, \ddot{q}, \mathbf{g}_0)$. The method operates in two recursive sweeps:

1. **Forward Recursion ($i = 1 \rightarrow 3$):** Propagates angular velocities $\boldsymbol{\omega}_i^i$, angular accelerations $\dot{\boldsymbol{\omega}}_i^i$, linear origin accelerations $\dot{\mathbf{p}}_i^i$, and center-of-mass (CoM) accelerations $\dot{\mathbf{p}}_{C_i}^i$ from the base to the end-effector.

2. **Backward Recursion ($i = 3 \rightarrow 1$):** Solves Newton translational balance ($\mathbf{f}_i^i - \mathbf{R}_{i+1}^i \mathbf{f}_{i+1}^{i+1} = m_i \dot{\mathbf{p}}_{C_i}^i$) and Euler rotational balance ($\boldsymbol{\mu}_i^i$) from the end-effector back to the base, projecting forces/torques onto joint motion axes $\mathbf{z}_0$ to obtain generalized actuation torques τ_i.

Why it is done: While the Lagrangian approach delivers closed-form equations of motion ($\mathbf{B}\ddot{q} + \mathbf{C}\dot{q} + \mathbf{g} = \boldsymbol{\tau}$) with $O(n^4)$ symbolic complexity, Newton-Euler vector recursion achieves $O(n)$ computational efficiency, making it the preferred method for real-time model-based control and dynamic simulation.

1.2 Forward & Backward Vector Equations

Forward Equations ($i = 1 \dots 3$): For revolute joints ($i = 2, 3$):

$$\boldsymbol{\omega}_i^i = (\mathbf{R}_i^{i-1})^T(\boldsymbol{\omega}_{i-1}^{i-1} + \dot{q}_i \mathbf{z}_0) \tag{1}$$
$$\dot{\boldsymbol{\omega}}_i^i = (\mathbf{R}_i^{i-1})^T(\dot{\boldsymbol{\omega}}_{i-1}^{i-1} + \ddot{q}_i \mathbf{z}_0 + \dot{q}_i \boldsymbol{\omega}_{i-1}^{i-1} \times \mathbf{z}_0) \tag{2}$$
$$\dot{\mathbf{p}}_i^i = (\mathbf{R}_i^{i-1})^T \dot{\mathbf{p}}_{i-1}^{i-1} + \dot{\boldsymbol{\omega}}_i^i \times \mathbf{r}_{i-1,i}^i + \boldsymbol{\omega}_i^i \times (\boldsymbol{\omega}_i^i \times \mathbf{r}_{i-1,i}^i) \tag{3}$$

For prismatic joint ($i = 1$):

$$\dot{\mathbf{p}}_1^1 = (\mathbf{R}_1^0)^T(\dot{\mathbf{p}}_0^0 - \mathbf{g}_0 + \ddot{q}_1 \mathbf{z}_0) \tag{4}$$

Center-of-Mass (CoM) linear acceleration for all links:

$$\dot{\mathbf{p}}_{C_i}^i = \dot{\mathbf{p}}_i^i + \dot{\boldsymbol{\omega}}_i^i \times \mathbf{r}_{i,C_i}^i + \boldsymbol{\omega}_i^i \times (\boldsymbol{\omega}_i^i \times \mathbf{r}_{i,C_i}^i) \tag{5}$$

Backward Equations ($i = 3 \dots 1$):

$$\mathbf{f}_i^i = \mathbf{R}_{i+1}^i \mathbf{f}_{i+1}^{i+1} + m_i \dot{\mathbf{p}}_{C_i}^i \tag{6}$$
$$\boldsymbol{\mu}_i^i = -\mathbf{f}_i^i \times (\mathbf{r}_{i-1,i}^i + \mathbf{r}_{i,C_i}^i) + \mathbf{R}_{i+1}^i \boldsymbol{\mu}_{i+1}^{i+1} + \mathbf{R}_{i+1}^i \mathbf{f}_{i+1}^{i+1} \times \mathbf{r}_{i,C_i}^i + \mathbf{I}_i^i \dot{\boldsymbol{\omega}}_i^i + \boldsymbol{\omega}_i^i \times (\mathbf{I}_i^i \boldsymbol{\omega}_i^i) \tag{7}$$

Joint Torque Projection:

$$\tau_i = \begin{cases} (\mathbf{f}_i^i)^T (\mathbf{R}_i^{i-1})^T \mathbf{z}_0 & \text{for Prismatic Joint (Joint 1)} \\ (\boldsymbol{\mu}_i^i)^T (\mathbf{R}_i^{i-1})^T \mathbf{z}_0 & \text{for Revolute Joints (Joints 2, 3)} \end{cases} \tag{8}$$

2 Linearity in Dynamic Parameters & Base Parameters

2.1 Linear Parameter Representation

What is being done: The dynamic model is expressed in linear regressor form:

$$\boldsymbol{\tau} = \mathbf{B}(q)\ddot{q} + \mathbf{C}(q, \dot{q})\dot{q} + \mathbf{g}(q) = \mathbf{Y}(q, \dot{q}, \ddot{q})\boldsymbol{\Theta} \tag{9}$$

Where $\boldsymbol{\Theta} \in \mathbb{R}^{30}$ contains 10 standard physical parameters per link ($m_i, m_i \mathbf{r}_{C_i}^i, \mathbf{I}_i^i$).

Why it is done: Linearity in dynamic parameters allows decoupling state kinematics ($q, \dot{q}, \ddot{q}$) into matrix $\mathbf{Y}$ while keeping unknown inertial parameters in vector $\boldsymbol{\Theta}$, making system identification solvable via linear least-squares.

2.2 Base Parameter Reduction ($\bar{\Theta}$)

What is being done: Not all 30 parameters independently affect joint torques. By eliminating unobservable parameters (zero columns in $\mathbf{Y}$) and grouping linearly dependent columns, we extract the minimal set of 5 base parameters $\bar{\boldsymbol{\Theta}} \in \mathbb{R}^5$:

$$\bar{\boldsymbol{\Theta}} = \begin{bmatrix} \bar{\theta}_1 \\ \bar{\theta}_2 \\ \bar{\theta}_3 \\ \bar{\theta}_4 \\ \bar{\theta}_5 \end{bmatrix} = \begin{bmatrix} m_1 + m_2 + m_3 \\ m_3 \\ I_{2yy} + I_{3yy} \\ I_{3xx} - I_{3yy} \\ I_{3zz} \end{bmatrix} = \begin{bmatrix} 3.000000 \text{ kg} \\ 1.000000 \text{ kg} \\ 0.026847 \text{ kg} \cdot \text{m}^2 \\ -0.013243 \text{ kg} \cdot \text{m}^2 \\ 0.013424 \text{ kg} \cdot \text{m}^2 \end{bmatrix} \tag{10}$$

The corresponding minimal base regressor matrix $\mathbf{X}(q, \dot{q}, \ddot{q}) \in \mathbb{R}^{3\times 5}$ is given by:

$$\mathbf{X}(q,\dot{q},\ddot{q}) = \begin{bmatrix} \ddot{q}_1 + g & 0.2\ddot{q}_3\cos q_3 - 0.2\dot{q}_3^2 \sin q_3 & 0 & 0 & 0 \\ 0 & \frac{\ddot{q}_2}{25}\cos^2 q_3 - \frac{\dot{q}_2\dot{q}_3}{25}\sin q_3 \cos q_3 & \ddot{q}_2 & \ddot{q}_2 \sin^2 q_3 + \dot{q}_2\dot{q}_3 \sin(2q_3) & 0 \\ 0 & 0.2\ddot{q}_1\cos q_3 + \frac{\ddot{q}_3}{25} - \frac{\dot{q}_2^2}{50}\sin(2q_3) + 0.2g\cos q_3 & 0 & -\frac{\dot{q}_2^2}{2}\sin(2q_3) & \ddot{q}_3 \end{bmatrix} \tag{11}$$

Why it is done: Reducing to base parameters guarantees full column rank for the stacked regressor matrix during system identification, preventing numerical ill-conditioning and matrix singularity.

3 Dynamic Parameter Identification via Least-Squares

3.1 Least-Squares (LS) Estimation Framework

What is being done: We collect $N = 500$ time samples along a persistently exciting sinusoidal trajectory ($q_1(t) = 0.5\sin t, q_2(t) = \cos(1.5t), q_3(t) = 0.8\sin(2t)$). Stacking $3N$ torque equations yields the linear system $\mathbf{W}\bar{\mathbf{\Theta}} = \mathbf{T}$, where $\mathbf{W} \in \mathbb{R}^{1500\times 5}$ and $\mathbf{T} \in \mathbb{R}^{1500}$. The optimal parameter estimates are solved via Moore-Penrose pseudo-inverse:

$$\hat{\bar{\mathbf{\Theta}}}_{LS} = (\mathbf{W}^T\mathbf{W})^{-1}\mathbf{W}^T\mathbf{T} \tag{12}$$

Why it is done: Least-Squares identification provides an optimal, noise-resilient estimate of the manipulator's base parameters required for adaptive and model-based feedback linearizing controllers.

3.2 Identification Results & Condition Number

Table 1 summarizes the numerical identification performance:

Table 1: Base Parameter Identification Results via Least-Squares.

Base Parameter	Symbolic Definition	Ground Truth	Estimated ($\hat{\bar{\theta}}_{LS}$)	Absolute Error
$\bar{\theta}_1$	$m_1 + m_2 + m_3$	3.000000	3.000000	$< 10^{-15}$
$\bar{\theta}_2$	m_3	1.000000	1.000000	0.00
$\bar{\theta}_3$	$I_{2yy} + I_{3yy}$	0.026847	0.026847	$< 10^{-15}$
$\bar{\theta}_4$	$I_{3xx} - I_{3yy}$	−0.013243	−0.013243	$< 10^{-15}$
$\bar{\theta}_5$	I_{3zz}	0.013424	0.013424	$< 10^{-15}$

The stacked regressor matrix condition number is $\text{cond}(\mathbf{W}) = 11.50$, proving high numerical stability and excellent trajectory excitation.